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Connection module

Abstract

A connection module is configured to be accommodated inside a battery pack including an upper case and a lower case, the connection module including: a plurality of bus bars that electrically connect a device disposed on the upper case side inside the battery pack to a connector disposed on the lower case side inside the battery pack; a plurality of holders that hold the plurality of bus bars, each of the plurality of holders including bus bar holding portions; and a bracket capable of locking the plurality of holders, wherein, in a state in which the plurality of holders that hold the plurality of bus bars are locked to the bracket, the plurality of bus bars are arranged side by side in a left-right direction orthogonal to a vertical direction, a direction orthogonal to the vertical direction and intersecting the left-right direction is assumed to be a front-rear direction.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a connection module.

BACKGROUND ART

(2) Some conventional casings such as an automobile battery pack are composed of an upper case and a lower case. A manufacturing process of such a battery pack or the like includes a step of electrically connecting devices mounted inside the casing, and a step of placing the upper case on the lower case so as to seal a gap therebetween. Designing of the casing may require connecting the devices disposed on the upper case side inside the casing to a connector for external output. In such a case, it is conceivable to provide the connector for external output on the upper case side as in the assembled battery described in JP 2019-186043A (Patent Document 1 below).

CITATION LIST

(3) Patent Document

(4) Patent Document 1: JP 2019-186043A

SUMMARY OF INVENTION

Technical Problem

(5) However, with the configuration as described above, the manufacturing process of the battery pack or the like cannot be completed even after finishing the step of sealing the gap between the upper case and the lower case after finishing the step of connecting all the devices inside the case (corresponding to a casing in the present application). The reason for this is that connection of the connector for external output needs to be performed separately after performing the step of sealing the gap between the upper case and the lower case. This may complicate the manufacturing process of the battery pack or the like.

(6) The present disclosure has been completed in light of the above-described circumstances, and an object thereof is to provide a connection module that allows a device disposed on the upper case side and a connector disposed on the lower case side inside a casing to be electrically connected to each other, and that can be easily attached.

Solution to Problem

(7) A connection module according to the present disclosure is a connection module configured to be accommodated inside a casing including an upper case and a lower case, the connection module including: a plurality of bus bars that electrically connect a device disposed on the upper case side inside the casing to a connector disposed on the lower case side inside the casing; a plurality of holders that hold the plurality of bus bars, each of the plurality of holders including a bus bar holding portion; and a bracket capable of locking the plurality of holders, wherein, in a state in which the plurality of holders that hold the plurality of bus bars are locked to the bracket, the plurality of bus bars are arranged side by side in a first direction orthogonal to a vertical direction, a direction orthogonal to the vertical direction and intersecting the first direction is assumed to be a second direction, the plurality of bus bars each include an upper connection portion extending in the second direction so as to be connected to the device, a lower connection portion extending in the second direction so as to be connected to the connector, and a coupling portion extending in the vertical direction so as to connect the upper connection portion and the lower connection portion to each other, a clearance is set between each of the bus bars and the corresponding holder in the second direction, and clearances are set between each of the holders and the bracket in the vertical direction and the first direction, respectively.

Advantageous Effects of Invention

(8) According to the present disclosure, it is possible to provide a connection module that allows a device disposed on the upper case side and a connector disposed on the lower case side inside a casing to be electrically connected to each other, and that can be easily attached.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a perspective view showing an example in which a connection module according to Embodiment 1 is applied to a battery pack.

(2) FIG. **2** is a side view of the connection module applied to the battery pack.

(3) FIG. **3** is a view showing the periphery of a right end holder in a cross section taken along the line A-A in FIG. **2**.

(4) FIG. **4** is a plan view of the connection module before a cover is fixed thereto.

(5) FIG. **5** is a perspective view of the connection module before the cover is fixed thereto.

(6) FIG. **6** is a cross-sectional view taken along the line B-B in FIG. **4**.

(7) FIG. **7** is a view showing the periphery of a holder in a cross section taken along the line C-C in FIG. **4**.

(8) FIG. **8** is an upper perspective view of the holder.

(9) FIG. **9** is a lower perspective view of the holder.

(10) FIG. **10** is a plan view of two holders that are coupled to each other.

(11) FIG. **11** is a cross-sectional view taken along the line D-D in FIG. **10**.

(12) FIG. **12** is a perspective view of bus bars.

(13) FIG. **13** is a perspective view of a bracket.

(14) FIG. **14** is a perspective view of a cover.

(15) FIG. **15** is a plan view showing locking between a lock receiving portion of a bracket and a lock portion of a holder according to Embodiment 2.

(16) FIG. **16** is a cross-sectional view taken along the line E-E in FIG. **15**.

(17) FIG. **17** is a cross-sectional view taken along the line F-F in FIG. **15**.

(18) FIG. **18** is a plan view of a bracket according to Embodiment 3.

(19) FIG. **19** is a plan view of a connection module before a cover is fixed thereto.

(20) FIG. **20** is a cross-sectional view taken along the line G-G in FIG. **19**.

(21) FIG. **21** is a view showing attachment between holders and the bracket in a cross section taken along the line G-G in FIG. **19**.

DESCRIPTION OF EMBODIMENTS

Description of Embodiments of the Present Disclosure

(22) First, aspects of the present disclosure will be enumerated and described below.

(23) (1) A connection module according to the present disclosure is a connection module configured to be accommodated inside a casing including an upper case and a lower case, the connection module including: a plurality of bus bars that electrically connect a device disposed on the upper case side inside the casing to a connector disposed on the lower case side inside the casing; a plurality of holders that hold the plurality of bus bars, each of the plurality of holders including a bus bar holding portion; and a bracket capable of locking the plurality of holders, wherein, in a state in which the plurality of holders that hold the plurality of bus bars are locked to the bracket, the plurality of bus bars are arranged side by side in a first direction orthogonal to a vertical direction, a direction orthogonal to the vertical direction and intersecting the first direction is assumed to be a second direction, the plurality of bus bars each include an upper connection portion extending in the second direction so as to be connected to the device, a lower connection portion extending in the second direction so as to be connected to the connector, and a coupling portion extending in the vertical direction so as to connect the upper connection portion and the lower connection portion to each other, a clearance is set between each of the bus bars and the corresponding holder in the second direction, and clearances are set between each of the holders and the bracket in the vertical direction and the first direction, respectively. Being "orthogonal" as used herein includes being orthogonal, and also includes being regarded as substantially orthogonal

such as intersecting at an angle of from about 85° to 95° even if not being orthogonal.
(24) With this configuration, the plurality of bus bars electrically connect the device disposed on the upper case side inside the casing to the connector disposed on the lower case side inside the casing. Accordingly, it is possible to complete the manufacturing process of a battery pack or the like by placing the upper case on the lower case so as to seal a gap therebetween after completing electrical connection of all the devices mounted inside the casing. In addition, the clearance is set between each of the bus bars and the corresponding holder in the front-rear direction, and the clearances are set between each of the holders and the bracket in the vertical direction and the left-right direction, respectively. Accordingly, it is possible to absorb positional shift and dimensional variation of the members and mounted devices in each of the directions.
(25) (2) It is preferable that each of the plurality of holders includes a lock portion and a lock receiving portion, and the adjacent holders of the plurality of holders are locked to each other with a clearance therebetween in the first direction by the lock portion of one of the holders and the lock receiving portion of the other holder being locked to each other.
(26) With this configuration, the holders each include the lock portion and the lock receiving portion, and it is therefore possible to couple the adjacent holders to each other, thus facilitating attachment between the holders. In addition, there is no need to separately form the holder that does not include the lock portion. Since the plurality of holders are locked to each other with the clearance therebetween in the left-right direction, it is possible to absorb positional shift and dimensional variation of the members and mounted devices in the left-right direction.
(27) (3) It is preferable that the plurality of holders include an intermediate holder and end holders, in a state in which the plurality of holders are locked to the bracket, the holders are disposed on opposite sides of the intermediate holder in the first direction, and the holder is disposed only one side of the end holders in the first direction, the intermediate holder includes both a lock portion and a lock receiving portion, one of the end holders includes at least the lock portion out of the lock portion and the lock receiving portion, the other end holder includes only the lock receiving portion out of the lock portion and the lock receiving portion, and the adjacent holders of the plurality of holders are locked to each other with a clearance therebetween in the first direction by the lock portion of one of the holders and the lock receiving portion of the other holder being locked to each other.
(28) With this configuration, it is possible to couple the adjacent holders to each other, thus facilitating attachment between the holders. In addition, the configuration includes the end holder that does not include the lock portion, and it is therefore possible to reduce the length of the bracket in the left-right direction. Since the plurality of holders are locked to each other with the clearance therebetween in the left-right direction, it is possible to absorb positional shift and dimensional variation of the members and mounted devices in the left-right direction.
(29) (4) It is preferable that, on each of the holder, a holder-side guide is provided in the vicinity of the lock portion, and a holder-side guide receiving portion is provided in the vicinity of the lock receiving portion, the holder-side guide protrudes outward in the first direction, and the holder-side guide receiving portion receives the holder-side guide.
(30) This configuration facilitates attachment between the holders.
(31) (5) It is preferable that the bracket is provided with a rail extending in the first direction, and each of the holders is provided with a rail-side guide configured to be fitted to the rail.
(32) This configuration facilitates attachment of the holders to the bracket.
(33) (6) It is preferable that a plurality of the rails and a plurality of the rail-side guides are provided so as to be spaced apart in the second direction.
(34) With this configuration, it is possible to increase the locking force of the holders provided by the bracket. In addition, it is possible to suppress rattling of the holders attached to the bracket.
(35) (7) It is preferable that each of the plurality of holders is provided with an engaging portion that protrudes downward, and an engagement receiving portion with which the engaging portion

engages from below is formed in the bracket.
(36) This configuration allows the holders to be attached to the bracket from above, thus improving the ease of attachment of the holders to the bracket.
(37) (8) It is preferable that the connection module further includes a cover configured to cover the bus bars and the holders from above and be fixed to the bracket.
(38) With this configuration, it is possible to prevent the bus bars and the holders from being detached from the bracket.
Details of Embodiments of the Present Disclosure
(39) Hereinafter, embodiments of the present disclosure will be described. It should be noted that the present disclosure is not limited to these examples, but is defined by the claims, and is intended to include all modifications which fall within the scope of the claims and the meaning and scope of equivalents thereof.
Embodiment 1
(40) Embodiment 1 of the present disclosure will be described with reference to FIGS. **1** to **14**. The following description will be given assuming that the direction indicated by the arrow Z is the upward direction, the direction indicated by the arrow X is the forward direction, and the direction indicated by the arrow Y is the leftward direction. Here, the left-right direction is an example of a first direction, and the front-rear direction is an example of a second direction. For a plurality of identical members, reference numerals may be assigned to some of the members, and reference numerals may be omitted for the other members.
(41) Connection Module
(42) As shown in FIG. **5**, a connection module **10** according to Embodiment 1 includes a plurality of bus bars **20**, a plurality of holders **30** that hold the plurality of bus bars **20**, and a bracket **50** that is locked to the plurality of holders **30**. As shown in FIG. **2**, the connection module **10** is accommodated inside a battery pack **1** (an example of a casing). As will be described below, the connection module **10** is placed on a terminal block **3** inside the battery pack **1**, and electrically connects a device-side connection portion **4**A of a device **4** (not shown) and a connector-side connection portion **7**B of a connector **5** to each other.
(43) Battery Pack, Upper Case, Lower Case
(44) As shown in FIG. **2**, the battery pack **1** includes an upper case **1**U that is open upward, and a lower case **1**L that is open downward. A known sealing member (not shown) is interposed between an edge portion of the opening of the upper case **1**U and an edge portion of the opening of the lower case **1**L. The upper case **1**U is placed on the lower case **1**L so as to close the openings, and a gap therebetween is sealed in a liquid-tight manner by the sealing member, whereby the battery pack **1** provides waterproofing.
(45) As shown in FIG. **1**, a battery cell B having a rectangular parallelepiped shape is placed inside the lower case **1**L. The upper, front, and rear sides of the battery cell B are covered by a battery bracket **2** formed in the shape of a protrusion that protrudes upward, and the battery cell B is fixed to the lower case **1**L. A terminal block **3** having a rectangular shape in a plan view is mounted and fixed to the battery bracket **2**. The terminal block **3** is formed to have a smaller width in the left-right direction than the width in the left-right direction of the battery bracket **2**. The terminal block **3** includes a rear terminal block **3**R that is mounted to an upper surface of the battery bracket **2** and that extends rearward, and a front terminal block **3**F that is mounted to a front surface of the battery bracket **2** and that extends downward and forward from the front end of the rear terminal block **3**R. As shown in FIG. **2**, the rear terminal block **3**R protrudes above an upper edge portion of the lower case **1**L in a state in which the rear terminal block **3**R is mounted to the lower case **1**L. Thus, in a state in which the lower case **1**L and the upper case **1**U are attached to each other, the rear terminal block **3**R is disposed on the upper case **1**U side inside the battery pack **1**.
(46) In a state in which the lower case **1**L and the upper case **1**U are attached to each other, the device **4** (not shown) is disposed on the upper case **1**U side inside the battery pack **1**, and the

device-side connection portion **4**A extends forward from the device **4** as shown in FIG. **1**. The device-side connection portion **4**A is provided rearward of the rear terminal block **3**R, and is held by at least one locking piece **3**B that protrudes upward. A bolt fixing portion **3**A that receives a bolt BT**1** is formed forward of the locking piece **3**B of the rear terminal block **3**R so as to protrude upward, and a bolt fastening hole (not shown) formed in a front end portion of the device-side connection portion **4**A is disposed on the bolt fixing portion **3**A. The connection module **10** is placed at a portion of the rear terminal block **3**R that is located forward of the bolt fixing portion **3**A, thus allowing each bus bar **20** of the connection module **10** to be connected to the corresponding device-side connection portion **4**A. As shown in FIG. **3**, a nut accommodating portion **3**C that accommodates a nut N that is fastened to a bolt BT**3** described below is provided at each of the right and left ends of the rear terminal block **3**R at a position at which the connection module **10** is placed. Although not shown, projection receiving portions that receive projecting portions **59** of a bracket **50** described below are provided forward of the right nut accommodating portion **3**C and rearward of the left nut accommodating portion **3**C, and the connection module **10** is fixed to the terminal block **3** by locking between the projecting portions **59** and the projection receiving portions.

(47) As shown in FIGS. **1** and **2**, a connector **5** for external output is disposed on the front side of the lower case **1**L. As shown in FIG. **1**, the connector **5** is formed extending through a side surface of the lower case **1**L on the front side in the front-rear direction. The connector **5** includes a hood portion **6** that is open forward, terminal fittings **7**, and a terminal accommodating portion **8** that extends rearward of the hood portion **6** and in which the terminal fittings **7** are accommodated side by side in the left-right direction. A rear end portion of the terminal accommodating portion **8** is fixed by the front terminal block **3**F. As shown in FIG. **2**, each terminal fitting **7** includes a tab **7**A that protrudes forward inside the hood portion **6**. A rear end portion of the terminal fitting **7** serves as a connector side connection portion **7**B, and a bolt fastening hole (not shown) is formed in the connector-side connection portion **7**B. As shown in FIG. **1**, the terminal accommodating portion **8** is open upward and rearward, and is configured to receive the bus bars **20** of the connection module **10** from above and connect the bus bars **20** to the corresponding connector-side connection portions **7**B. Bolt fixing portions (not shown) are provided at a rear end portion of the terminal accommodating portion **8** so as to protrude upward and receive bolts BT**2**. Bolt fastening holes (not shown) are disposed on the bolt fixing portions.

(48) Bus Bar, Upper Connection Portion, Lower Connection Portion, Coupling Portion

(49) Each bus bar **20** is made of a conductive metal such as copper, and has the shape of an elongated plate that is bent in a crank shape as shown in FIG. **12**. The bus bar **20** includes an upper connection portion **21**, a lower connection portion **22**, and a coupling portion **23** that connects the upper connection portion **21** and the lower connection portion **22** to each other. The coupling portion **23** extends in a vertical direction, and the front-rear direction thereof corresponds to the plate thickness direction. The upper connection portion **21** extends rearward continuously from the upper end of the coupling portion **23**, and the vertical direction thereof corresponds to the plate thickness direction. The lower connection portion **22** extends forward continuously from the lower end of the coupling portion **23**, and the vertical direction thereof corresponds to the plate thickness direction. Bolt fastening holes **21**A and **22**A are respectively formed on the rear side of the upper connection portion **21** and the front side of the lower connection portion **22**. A pair of recesses **24** are provided at left and right end portions of the upper connection portion **21**. The recesses **24** are disposed on the front side of the upper connection portion **21**. The dimensions of each bus bar **20** in the plate thickness direction and the width direction can be changed according to the width of a complementary member to which the bus bar **20** is to be connected, the desired strength, and so forth. The plurality of (eight in the present embodiment) bus bars **20** may have the same shape, or may have different shapes. Three types of bus bars that differ in dimensions are used as the bus bars **20** according to the present embodiment.

(50) Holder, Intermediate Holder, Right End Holder, Left End Holder

(51) The plurality of holders **30** are composed of intermediate holders **30**M, a right end holder **30**R (an example of an end holder), and a left end holder **30**L (an example of an end holder). Although the details will be described below, the plurality of holders **30** are coupled to each other in the left-right direction in the connection module **10** as shown in FIG. **5**. The right end holder **30**R is disposed at the right end, and the left end holder **30**L is disposed at the left end. The intermediate holders **30**M are disposed between the right end holder **30**R and the left end holder **30**L. In the following, when there is no need to distinguish between the intermediate holders **30**M, the right end holder **30**R, and the left end holder **30**L, the term holder **30** will be used.

(52) The holder **30** is made of an insulating synthetic resin, and has the shape of a rectangular frame that is open upward as shown in FIG. **8**. One holder **30** is configured to hold one bus bar **20**. Each holder **30** is formed so as to correspond to the dimensions of the corresponding bus bar **20**. In the present embodiment, three types of holders that differ in dimensions are used as the holders **30** that hold the bus bars **20** according to the dimensions of the bus bars **20**. Each holder **30** has a pair of side walls **31**A and **31**B in the left-right direction. The pair of side walls **31**A and **31**B are connected to each other by a front wall **32** at the front end, and by a rear wall **33** at the rear end. The lower ends of the pair of side walls **31**A and **31**B, the front wall **32**, and the rear wall **33** are connected to each other by the bottom wall **34**.

(53) Bus Bar Holding Portion, Rail-Side Guide

(54) As shown in FIG. **8**, the holder **30** includes a pair of bus bar holding portions **35**A and **35**B on the pair of side walls **31**A and **31**B. The bus bar holding portions **35**A and **35**B are disposed at a central portion of the holder **30** in the front-rear direction. The bus bar holding portions **35**A and **35**B are formed protruding toward the respective opposing side walls **31**B and **31**A. Cut-out portions **36** are formed in portions of the side walls **31**B and **31**A that are located forward and rearward of the bus bar holding portions **35**A and **35**B. Thus, central portions of the side walls **31**A and **31**B respectively including the bus bar holding portions **35**A and **35**B can undergo bending deformation in the left-right direction. Tapered surfaces **37** that are inclined so as to respectively approach the side walls **31**A and **31**B in a direction toward the upper side are provided on the upper side of the bus bar holding portions **35**A and **35**B. As shown in FIG. **6**, each of the lower ends of the bus bar holding portions **35**A and **35**B forms a lower surface **35**C having a steep shape. As shown in FIG. **7**, upper surfaces **32**A and **33**A of the front wall **32** and the rear wall **33** are provided below the lower surfaces **35**C of the bus bar holding portions **35**A and **35**B, and the interval in the vertical direction between the lower surfaces **35**C of the bus bar holding portions **35**A and **35**B and the upper surfaces **32**A and **33**A of the front wall **32** and the rear wall **33** is set to be the same as or slightly larger than the thickness of the upper connection portion **21**. As shown in FIG. **8**, a pair of protrusions **38** are provided at left and right end portions of the front wall **32** so as to be continuous from the pair of side walls **31**A and **31**B. The protrusions **38** are disposed above the upper surface **32**A of the front wall **32**. As shown in FIG. **4**, the thickness of each protrusion **38** in the front-rear direction is set to be smaller than the length of each recess **24** in the front-rear direction. As shown in FIG. **9**, a pair of rail-side guides **39** are provided on the front and rear sides of the bottom wall **34** so as to extend downward. The lower ends of the pair of rail-side guides **39** extend in the front-rear direction so as be away from each other.

(55) Lock Portion, First Holder-Side Guide, Second Holder-Side Guide

(56) As shown in FIG. **8**, the intermediate holder **30**M includes, on the side wall **31**A, a lock piece **40**, at least one first holder-side guide **41** (an example of a holder-side guide), and at least one second holder-side guide **42** (an example of the holder side guide). The lock piece **40** is disposed at a central portion of each intermediate holder **30**M in the front-rear direction, and protrudes leftward of the side wall **31**A. A lock portion **40**A is provided at a distal end of the lock piece **40** so as to protrude upward. The left side of the lock portion **40**A forms a tapered surface **40**B that extends upward in a direction toward the right side. The right end of the lock portion **40**A has a steep shape.

The first holder-side guide **41** and the second holder-side guide **42** protrude leftward of the side wall **31**A, and the protruding lengths thereof are smaller than the protruding length of the lock piece **40**. A pair of second holder-side guides **42** are provided on the front and rear sides of the side wall **31**A. The second holder-side guides **42** have a length in the front-rear direction that is larger than a length thereof in the vertical direction. A pair of first holder-side guides **41** are disposed between the lock piece **40** and the second holder-side guides **42** in the front-rear direction. The first holder-side guides **41** have a length in the vertical direction that is larger than a length thereof in the front-rear direction. As in the case of the intermediate holder **30**M, the right end holder **30**R includes a lock piece **40**, at least one first holder-side guide **41**, and at least one second holder-side guide **42**.
(57) Lock Receiving Portion, First Holder-Side Guide Receiving Portion, Second Holder-Side Guide Receiving Portion
(58) As shown in FIG. **8**, the bottom wall **34** of the holder **30** has a lock receiving portion **43** formed in a central portion thereof in the front-rear direction. The lock receiving portion **43** is located on the right side of the bottom wall **34**. As shown in FIG. **6**, the lock receiving portion **43** is configured to be locked to a lock portion **40**A of another holder **30** that is adjacent on the right side. A clearance CL**4** in the left-right direction is set for the locking between the lock portion **40**A and the lock receiving portion **43**. As shown in FIG. **8**, a pair of second holder-side guide receiving portions **44** (an example of a holder-side guide receiving portion) are provided on the front and rear sides on the side wall **31**B side of the bottom wall **34**. The second holder-side guide receiving portions **44** are formed so as to be recessed below the height of the bottom wall **34** on which the lock receiving portions **43** are located. The pair of second holder-side guide receiving portions **44** are connected to the front wall **32** or the rear wall **33**, but not connected to the side wall **31**B, and are open rightward. Accordingly, as shown in FIG. **11**, the pair of second holder-side guide receiving portions **44** are configured to receive a pair of second holder-side guides **42** of another holder **30** that is adjacent on the right side. The first holder-side guides **41** can be engaged with first holder-side guide receiving portions **45** (an example of a holder-side guide receiving portion) that are each formed by a portion of the bottom wall **34**.
(59) As described above, the plurality of holders **30** are attached to the connection module **10** while being coupled to each other in the left-right direction by locking between the lock portion **40**A and the lock receiving portion **43**. At this time, the holders **30** are disposed on opposite sides of each intermediate holder **30**M in the left-right direction. On the other hand, as shown in FIGS. **6** and **10**, no holder **30** is disposed on the left side of the end holder **30**L, and therefore the end holder **30**L is not provided with the lock piece **40** including the lock portion **40**A, the first holder-side guides **41**, and the second holder-side guide **42**. Similarly, as shown in FIG. **3**, no holder **30** is disposed on the right side of the right end holder **30**R. Therefore, the right end holder **30**R may or may not include the lock receiving portion **43**, the first holder-side guide receiving portions **45**, and the second holder-side guide receiving portions **44**. Note that, in Embodiment 1, the intermediate holder **30**M including the lock receiving portion **43**, the first holder-side guide receiving portions **45**, and the second holder-side guide receiving portions **44** is also used as the right end holder **30**R.
(60) Bracket, Rail
(61) The bracket **50** is made of an insulating synthetic resin, and includes, as shown in FIG. **13**, a holder base portion **51** and a bus bar accommodating portion **52**. The holder base portion **51** has the shape of a frame that extends in the left-right direction. A pair of rails **53** extending through the holder base portion **51** in the vertical direction are formed in a slit shape on the holder base portion **51** so as to extend in the left-right direction. The pair of rails **53** are arranged side by side and spaced apart in the front-rear direction. As shown in FIG. **4**, the right end of each rail **53** serves as an insertion opening **54** that is formed to be wide. As shown in FIG. **13**, a stopper portion **55** in the form of an upwardly extending wall surface is provided at the left ends of the rails **53**. As shown in FIG. **6**, in a state in which the plurality of holders **30** are attached to the bracket **50**, the stopper

portion **55** faces the side wall **31**A of the left end holder **30**L. As shown in FIG. **13**, the pair of rails **53** includes a rear rail **53**B located on the rear side and a front rail **53**F located on the front side. A rear extension portion **56**B that extends forward is formed at a rear edge portion of the rear rail **53**B. A front extension portion **56**F that extends rearward is formed at a front hole edge portion of the front rail **53**F. The rear extension portion **56**B and the front extension portion **56**F are collectively denoted as the extension portion **56**. As shown in FIG. **7**, the front extension portion **56**F and the rear extension portion **56**B support the holders **30** from below in a state in which the holders **30** are attached to the bracket **50**. As shown in FIG. **4**, fixing holes **57** for fixing a cover **60** described below are formed in the left and right ends of the holder base portion **51**. Projection receiving portions **59**B are provided rearward of the right fixing hole **57** and forward of the left fixing hole **57**. The projection receiving portions **59**B are configured to receive projecting portions **67** of a cover **60** described below. As shown in FIG. **13**, projecting portions **59**A are provided forward of the right fixing hole **57** and rearward of the left fixing hole **57** so as to protrude downward. The bus bar accommodating portion **52** is provided extending downward from the front end of the holder base portion **51**. The bus bar accommodating portion **52** includes a plurality of partition plates **58** that are arranged side by side in the left-right direction. As shown in FIG. **5**, the plurality of holders **30** and the plurality of bus bars **20** are attached to the bracket **50**, and the bus bars **20** are separated by the partition plates **58** so as not to be in contact with each other.

(62) Cover

(63) The cover **60** is made of an insulating synthetic resin, and includes, as shown in FIG. **14**, a body portion **61** having a plate shape that extends in the left-right direction, and fixing portions **66** disposed on opposite sides of the body portion **61** in the left-right direction. The fixing portions **66** each have a shape that protrudes downward as compared with the body portion **61**. A fixing hole **62** is formed in a central portion of each of the fixing portions **66** in the front-rear direction. Projecting portions **67** are provided rearward of the right fixing hole **62** and forward of the left fixing hole **62** so as to protrude downward. A protection wall **63** is provided at the front end of the body portion **61** so as to extend downward. Groove portions **64** are formed in the protection wall **63**. As shown in FIGS. **1** and **3**, the cover **60** is placed on and fixed to the bracket **50** to which the plurality of holders **30** and the plurality of bus bars **20** are attached. As shown in FIG. **1**, the groove portions **64** are configured to receive the partition plates **58**. As shown in FIG. **3**, the right end of the body portion **61** serves as a retaining portion **65** in the form of a wall surface that extends downward, and faces the side wall **31**B of the right end holder **30**R. The retaining portion **65** prevents the plurality of holders **30** from being dislodged to the right of the bracket **50**.

(64) Attachment of Connection Module

(65) By inserting the rail-side guides **39** of each holder **30** into the insertion opening **54** of the bracket **50**, and moving the holder **30** to the left, the holder **30** is attached to the bracket **50** in a state in which the rail-side guide **39** are fitted to the rails **53** (see FIG. **4**). At this time, the left end holder **30**L is attached first, and the right end holder **30**R is attached last. In the process of this attachment, adjacent holders **30** are coupled to each other in the left-right direction. When holders **30** are moved toward each other, the tapered surface **40**B of a holder **30** comes into contact with the side wall **31**B of a holder **30** adjacent on the left side. By moving the holders **30** further toward each other, the lock piece **40** enters below the bottom wall **34** of the holder **30** on the left side while being bent downward, and is locked to the lock receiving portion **43** (see FIG. **6**). During this operation, the first holder-side guides **41** and the second holder-side guides **42** of the holder **30** are respectively engaged with the first holder-side guide receiving portions **45** and the second holder-side guide receiving portions **44** of the holder **30** on the left side, thus assisting in locking between the lock portion **40**A and the lock receiving portion **43** (see FIGS. **10** and **11**). In a state in which the holders **30** are attached to the bracket **50**, a clearance CL**2** is set between each of the holders **30** and the bracket **50** in the vertical direction as shown in FIG. **7**, and a clearance CL**3** is set between each of the holders **30** and the bracket **50** in the left-right direction as shown in FIG. **6**.

(66) Next, each bus bar **20** is attached to the corresponding holder **30**. The upper connection portion **21** is pressed against the holder **30** from above, and comes into contact with the tapered surfaces **37** of the bus bar holding portions **35**A and **35**B. The upper connection portion **21** slides on the tapered surfaces **37** while bending the bus bar holding portions **35**A and **35**B outward, and moves over the bus bar holding portions **35**A and **35**B (see FIG. **6**). In a state in which the bus bar **20** is held by the holder **30**, the bus bar **20** is located below the lower surfaces **35**C of the bus bar holding portions **35**A and **35**B. At this time, as shown in FIG. **4**, the bus bar **20** is positioned relative to the holder **30** such that the protrusions **38** of the holder **30** are fitted to the recesses **24** of the upper connection portion **21**. A clearance CL**1** is set between the holder **30** and the bus bar **20** in the front-rear direction.
(67) As shown in FIG. **3**, finally the cover **60** is placed on the bracket **50** from above, and the projecting portions **67** and the projection receiving portion **59**B are locked to each other, whereby the attachment of the connection module **10** is completed.
(68) Mounting of Connection Module to Battery Pack
(69) As shown in FIGS. **1** and **2**, the connection module **10** that has been attached in the above-described manner is placed on the terminal block **3** inside the battery pack **1**, and the connection module **10** is fixed to the terminal block **3** by locking between the projecting portions **59** of the bracket **50** and projection receiving portions (not shown) of the terminal block **3**. Bolts BT**3** are inserted into the fixing holes **62** and **57**, and nuts N and the bolts BT**3** are fastened together, thus further fixing the cover **60** to the bracket **50**. Bolts BT**1** are inserted into the bolt fastening holes **21**A of the upper connection portions **21** and bolt fastening holes (not shown) of the device-side connection portions **4**A, and the bolts BT**3** are fastened to the bolt fixing portions **3**A, whereby the connection module **10** and the device **4** are electrically connected to each other. Bolts BT**2** are inserted into the bolt fastening holes **22**A of the lower connection portions **22** and bolt fastening holes (not shown) of the connector-side connection portions **7**B, and the bolts BT**2** are fastened to the bolt fixing portions of the terminal accommodating portion **8**, whereby the connection module **10** and the connector **5** are electrically connected to each other. Accordingly, the device **4** and the connector **5** are electrically connected to each other.
(70) Operations and Effects of Embodiment 1
(71) According to Embodiment 1, the following operations and effects are achieved.
(72) The connection module **10** according to Embodiment 1 is a connection module **10** configured to be accommodated inside a battery pack **1** including an upper case **1**U and a lower case **1**L, the connection module including: a plurality of bus bars **20** that electrically connect a device **4** disposed on the upper case **1**U side inside the battery pack **1** to a connector **5** disposed on the lower case **1**L side inside the battery pack **1**; a plurality of holders **30** that hold the plurality of bus bars **20**, each of the plurality of holders **30** including bus bar holding portions **35**A and **35**B; and a bracket **50** capable of locking the plurality of holders **30**, wherein, in a state in which the plurality of holders **30** that hold the plurality of bus bars **20** are locked to the bracket **50**, the plurality of bus bars **20** are arranged side by side in a left-right direction orthogonal to a vertical direction, a direction orthogonal to the vertical direction and intersecting the left-right direction is assumed to be a front-rear direction, the plurality of bus bars **20** each include an upper connection portion **21** extending in the front-rear direction so as to be connected to the device **4**, a lower connection portion **22** extending in the front-rear direction so as to be connected to the connector **5**, and a coupling portion **23** extending in the vertical direction so as to connect the upper connection portion **21** and the lower connection portion **22** to each other, a clearance CL**1** is set between each of the bus bars **20** and the corresponding holder in the front-rear direction, and a clearance CL**2** and a clearance CL**3** are set between each of the holders **30** and the bracket **50** in the vertical direction and the left-right direction, respectively.
(73) With the above-described configuration, the plurality of bus bars **20** electrically connect the device **4** disposed on the upper case **1**U side inside the battery pack **1** to the connector **5** disposed

on the lower case **1**L side inside the battery pack **1**. Accordingly, it is possible to complete the manufacturing process of the battery pack **1** by placing the upper case **1**U on the lower case **1**L so as to seal a gap therebetween after completing electrical connection of all the devices mounted inside the battery pack **1**. In addition, the clearance CL**1** is set between each of the bus bars **20** and the corresponding holder **30** in the front-rear direction, and the clearance CL**2** and the clearance CL**3** are set between each of the holders **30** and the bracket **50** in the vertical direction and the left-right direction, respectively. Accordingly, it is possible to absorb positional shift and dimensional variation of the members and mounted devices in each of the directions.
(74) In Embodiment 1, the plurality of holders **30** include an intermediate holder **30**M, a right end holder **30**R, and a left end holder **30**L. In a state in which the plurality of holders **30** are locked to the bracket **50**, the holders **30** are disposed on opposite sides of the intermediate holder **30**M in the left-right direction. The holders **30** are disposed only on the left side of the right end holder **30**R in the left-right direction. The holders **30** are disposed only on the right side of the left end holder **30**L in the left-right direction. The intermediate holder **30**M includes both a lock portion **40**A and a lock receiving portion **43**. The right end holder **30**R includes the lock portion **40**A and the lock receiving portion **43**. The left end holder **30**L includes only the lock receiving portion **43** out of the lock portion **40**A and the lock receiving portion **43**. The adjacent holders **30** of the plurality of holders **30** are locked to each other with a clearance CL**4** therebetween in the left-right direction by the lock portion **40**A of one of the holders **30** and the lock receiving portion **43** of the other holder **30** being locked to each other.
(75) With the above-described configuration, it is possible to couple the adjacent holders **30** to each other, thus facilitating attachment between the holders **30**. In addition, the configuration includes the left end holder **30**L that does not include the lock portion **40**A, and it is therefore possible to reduce the length of the bracket **50** in the left-right direction. Since the plurality of holders **30** are locked to each other with the clearance CL**4** therebetween in the left-right direction, it is possible to absorb positional shift and dimensional variation of the members and mounted devices in the left-right direction.
(76) In Embodiment 1, on each of the holders **30**, a first holder-side guide **41** and a second holder-side guide **42** are provided in the vicinity of the lock portion **40**A, a first holder-side guide receiving portion **45** and a second holder-side guide receiving portion **44** are provided in the vicinity of the lock receiving portion **43**, the first holder-side guide **41** and the second holder-side guide **42** protrude outward in the left-right direction, and the first holder-side guide receiving portion **45** and the second holder-side guide receiving portion **44** receive the first holder-side guide **41** and the second holder-side guide **42**.
(77) The above-described configuration facilitates attachment between the holders **30**.
(78) In Embodiment 1, the bracket **50** is provided with a rail **53** extending in the left-right direction, and each of the holders **30** is provided with a rail-side guide **39** configured to be fitted to the rails **53**.
(79) The above-described configuration facilitates attachment of the holders **30** to the bracket **50**.
(80) In Embodiment 1, a pair of the rails **53** and a pair of the rail-side guides **39** are provided so as to be spaced apart in the front-rear direction.
(81) With the above-described configuration, it is possible to increase the locking force of the holders **30** provided by the bracket **50**. In addition, it is possible to suppress rattling of the holders **30** attached to the bracket **50**.
(82) In Embodiment 1, the connection module **10** further includes a cover **60** configured to cover the bus bars **20** and the holders **30** from above and be fixed to the bracket **50**.
(83) With the above-described configuration, it is possible to prevent the bus bars **20** and the holders **30** from being detached from the bracket **50**.
Embodiment 2
(84) Embodiment 2 of the present disclosure will be described with reference to FIGS. **15** to **17**. In

the following, descriptions of the members, operations and effects that are the same as those of Embodiment 1 will be omitted. For a plurality of the same members, reference numerals may be assigned to some of the members, and reference numerals may be omitted for the other members.
(85) A connection module **110** according to Embodiment 2 includes a plurality of bus bars **20**, a plurality of holders **30** that hold the plurality of bus bars **20**, and a bracket **150** configured to be locked to the plurality of holders **30**.
(86) As shown in FIGS. **15** and **16**, a stopper portion **155** in the form of an upwardly extending wall surface is provided at the left end of the bracket **150**. A lock receiving portion **143** similar to the counterpart of the holder **30** is provided leftward and downward of the stopper portion **155**. The lock receiving portion **143** is locked to the lock portion **40**A of the holder **30**. As shown in FIG. **17**, the bracket **150** includes first holder-side guide receiving portions **145** and second holder-side guide receiving portions **144** similar to the counterparts of the holder **30**. The first holder-side guide receiving portions **145** and the second holder-side guide receiving portions **144** are respectively engaged with the first holder-side guides **41** and the second holder side guides **42** of the holder **30**.
(87) Operations and Effects of Embodiment 2
(88) According to Embodiment 2, the following operations and effects are achieved.
(89) As shown in FIG. **16**, each of the plurality of holders **30** includes a lock portion **40**A and a lock receiving portion **43**, and the adjacent holders **30** of the plurality of holders **30** are locked to each other with a clearance CL**4** therebetween in the left-right direction by the lock portion **40**A of one holder **30** and the lock receiving portion **43** of the other holder **30** being locked to each other.
(90) With the above-described configuration, the holders **30** each include a lock portion **40**A and a lock receiving portion **43**, and it is therefore possible to couple the adjacent holders **30** to each other, thus facilitating attachment between the holders **30**. In addition, there is no need to separately form the holder **30** that does not include the lock portion **40**A, such as the left end holder **30**L of Embodiment 1. Since the plurality of holders **30** are locked to each other with the clearance CL**4** therebetween in the left-right direction, it is possible to absorb positional shift and dimensional variation of the members and mounted devices in the left-right direction.
Embodiment 3
(91) Embodiment 3 of the present disclosure will be described with reference to FIGS. **18** to **21**. In the following, descriptions of the members, operations and effects that are the same as those of Embodiment 1 will be omitted. For a plurality of identical members, reference numerals may be assigned to some of the members, and reference numerals may be omitted for the other members.
(92) A connection module **210** according to Embodiment 3 includes a plurality of bus bars **20**, a plurality of holders **30** that hold the plurality of bus bars **20**, and a bracket **250** configured to be locked to the plurality of holders **30**.
(93) Operations and Effects of Embodiment 3
(94) As shown in FIG. **18**, a plurality of insertion openings **254** of the bracket **250** are formed in an extension portion **256**. As shown in FIG. **20**, each of the insertion openings **254** is disposed on the right side of a normal locking position of the corresponding holder **30**. As a result of the rail-side guide **39** (an example of an engaging portion) being engaged, from below, with engagement receiving portions **256**A of the extension portion **256** that are located leftward of the corresponding insertion openings **254**, the holders **30** are held so as to be prevented from being dislodged upward. Accordingly, it is possible to attach a plurality of holders **30** and the bracket **250** to each other by inserting the rail-side guides **39** of each of a plurality of previously coupled holders **30** into the corresponding insertion openings **254** from above, and thereafter moving the plurality of coupled holders **30** leftward as shown in FIG. **21**, instead of attaching the holders **30** one by one to the bracket **50** by inserting the rail-side guides **39** of each of the holders **30** from the insertion opening **54** located at the right end of the bracket **50** as in the case of Embodiment 1 (see FIG. **4**).
(95) Although the insertion opening **54** is provided at the right end of the bracket **50** (see FIG. **4**) in Embodiment 1, the insertion openings **254** of the connection module **210** are provided in the

extension portion **256** as shown in FIG. **19**. Accordingly, the bracket **250** can be formed smaller than the bracket **50** in the left-right direction.

Other Embodiments

(96) (1) Although Embodiments 1 to 3 have described cases where the connection modules **10**, **110**, and **210** are applied to the battery pack **1**, the present disclosure is not limited thereto. The connection modules are also applicable to casings other than a battery pack.

(97) (2) Although Embodiments 1 to 3 use configurations in which the bus bar accommodating portion **52** that accommodates the coupling portions **23** of the bus bars **20** are provided in each of the brackets **50**, **150**, and **250**, the present disclosure is not limited thereto. It is possible to adopt a configuration in which a bus bar accommodating portion for one holder is provided in each of the holders.

(98) 3) Although Embodiments 1 to 3 use configurations in which electrical connection between the bus bars **20** and the device **4**, and electrical connection between the bus bars **20** and the connector **5** are achieved through bolt fastening, the present disclosure is not limited thereto. It is possible to adopt a configuration in which these electrical connections are achieved through welding, for example.

LIST OF REFERENCE NUMERALS

(99) **1**: Battery pack **1**L: Lower case **1**U: Upper case **2**: Battery bracket **3**: Terminal block **3**A: Bolt fixing portion **3**C: Nut accommodating portion **3**B: Locking piece **59**B: Projection receiving portion **3**F: Front terminal block **3**R: Rear terminal block **4**: Device **25** **4**A: Device-side connection portion **5**: Connector **6**: Hood portion **7**: Terminal fitting **7**A: Tab **7**B: Connector-side connection portion **8**: Terminal accommodating portion **10**, **110**, **210**: Connection module **20**: Bus bar **21**: Upper connection portion **22**: Lower connection portion **21**A, **22**A: Bolt fastening hole **23**: Coupling portion **24**: Recess **30**: Holder **30**L: Left end holder **30**M: Intermediate holder **30**R: Right end holder **31**A, **31**B: Side wall **32**: Front wall **32**A, **33**A: Upper surface **33**: Rear wall **34**: Bottom wall **35**A, **35**B: Bus bar holding portion **35**C: Lower surface **36**: Cut-out portion **37**, **40**B: Tapered surface **38**: Protrusion **39**: Rail-side guide (example of engaging portion) **40**: Lock piece **40**A: Lock portion **41**: First holder-side guide **42**: Second holder-side guide **43**, **143**: Lock receiving portion **44**, **144**: Second holder-side guide receiving portion **45**, **145**: First holder-side guide receiving portion **50**, **150**, **250**: Bracket **51**: Holder base portion **52**: Bus bar accommodating portion **53**: Rail **53**F: Front rail **53**B: Rear rail **54**, **254**: Insertion opening **55**, **155**: Stopper portion **56**, **256**: Extension portion **56**F: Front extension portion **56**B: Rear extension portion **57**, **62**: Fixing hole **58**: Partition plate **59**A, **67**: Projecting portion **60**: Cover **61**: Body portion **63**: Protection wall **64**: Groove portion **65**: Retaining portion **66**: Fixing portion **256**A: Engagement receiving portion B: Battery cell BT**1**, BT**2**, BT**3**: Bolt CL**1**, CL**2**, CL**3**, CL**4**: Clearance N: Nut

Claims

1. A connection module configured to be accommodated inside a casing including an upper case and a lower case, the connection module comprising: a plurality of bus bars that electrically connect a device disposed on the upper case side inside the casing to a connector disposed on the lower case side inside the casing; a plurality of holders that hold the plurality of bus bars, each of the plurality of holders including a bus bar holding portion; and a bracket capable of locking the plurality of holders, wherein, in a state in which the plurality of holders that hold the plurality of bus bars are locked to the bracket, the plurality of bus bars are arranged side by side in a first direction orthogonal to a vertical direction, a direction orthogonal to the vertical direction and intersecting the first direction is assumed to be a second direction, the plurality of bus bars each include an upper connection portion extending in the second direction so as to be connected to the device, a lower connection portion extending in the second direction so as to be connected to the connector, and a coupling portion extending in the vertical direction so as to connect the upper

connection portion and the lower connection portion to each other, a clearance is set between each of the bus bars and the corresponding holder in the second direction, and clearances are set between each of the holders and the bracket in the vertical direction and the first direction, respectively.
2. The connection module according to claim 1, wherein each of the plurality of holders includes a lock portion and a lock receiving portion, and adjacent holders of the plurality of holders are locked to each other with a clearance therebetween in the first direction by the lock portion of one of the holders and the lock receiving portion of another holder being locked to each other.
3. The connection module according to claim 2, wherein, on each of the holders, a holder-side guide is provided in the vicinity of the lock portion, and a holder-side guide receiving portion is provided in the vicinity of the lock receiving portion, the holder-side guide protrudes outward in the first direction, and the holder-side guide receiving portion receives the holder-side guide.
4. The connection module according to claim 1, wherein the plurality of holders include an intermediate holder and end holders, in a state in which the plurality of holders are locked to the bracket, the holders are disposed on opposite sides of the intermediate holder in the first direction, and the holders are disposed on only one side of the end holders in the first direction, the intermediate holder includes both a lock portion and a lock receiving portion, one of the end holders includes at least a lock portion, another end holder includes only a lock receiving portion, and adjacent holders of the plurality of holders are locked to each other with a clearance therebetween in the first direction by a lock portion of one of the holders and a lock receiving portion of another holder being locked to each other.
5. The connection module according to claim 1, wherein the bracket is provided with a rail extending in the first direction, and each of the holders is provided with a rail-side guide configured to be fitted to the rail.
6. The connection module according to claim 5, wherein a plurality of the rails and a plurality of the rail-side guides are provided so as to be spaced apart in the second direction.
7. The connection module according to claim 1, wherein each of the plurality of holders is provided with an engaging portion that protrudes downward, and an engagement receiving portion with which the engaging portion engages from below is formed in the bracket.
8. The connection module according to claim 1, further comprising a cover configured to cover the bus bars and the holders from above and be fixed to the bracket.